

# EpiGraph: Dynamic Usage-Weighted Topology and Synaptic Consolidation for Multi-Hop Agentic Memory

Bhavyateja Potineni\*, Lohit Giri, Vadim Kutsyy, and Rajasekhara Pentakota

Independent Researchers

bhavyateja.potineni@gmail.com, lohit.giri@gmail.com, vadim@kutsyy.com, rajsekhar.cs@gmail.com

**Abstract**—As autonomous Large Language Model (LLM) agents are deployed across sustained, multi-session environments, conventional memory architectures suffer from three fundamental pathologies: (1) Associative Blindness, wherein flat dense vector indexes fail to traverse multi-hop relational dependencies across distant interaction sessions; (2) Scaffolding Amnesia, where standard wall-clock temporal decay rules aggressively erase foundational user identity and architectural constraints; and (3) Static Topology Stagnation, where existing graph-augmented retrieval systems treat network structure as immutable, ignoring real-time usage dynamics and task utility. In this work, we propose EpiGraph, a hierarchical hybrid memory architecture for autonomous agents grounded in Complementary Learning Systems (CLS) principles. EpiGraph partitions memory operations into a low-latency streaming ingestion and retrieval reflex (“Waking State”) and an asynchronous background synaptic consolidation cycle (“Dreaming State”). EpiGraph introduces Usage-Modulated Personalized PageRank (U-PPR), a spreading activation algorithm wherein transition probabilities adapt via activity-dependent Hebbian plasticity to dynamically promote persistent foundational entities into high-centrality Epistemic Macro-Hubs; Consolidation-Activated Topology Decay (CATD), a selective forgetting rule executed strictly off the read path that scales retention half-life by topological load-bearing weight rather than wall-clock recency, protected by a cold-start grace period ( $N_{\text{grace}} \geq 4$ ); and hierarchical context engineering fusing dense vectors, U-PPR graph traversal, and BM25 via dynamic intent-routed Reciprocal Rank Fusion (RRF). We evaluate EpiGraph locally across all 10 long-term multi-session conversations (1,982 evaluation QA pairs) in the LoCoMo benchmark. EpiGraph achieves a +38.9% relative improvement in Recall@5 (53.21% vs. 38.29%, paired  $t = 13.57$ ,  $p < 0.001$ ) and a +43.1% improvement in MRR (0.4203 vs. 0.2937,  $p < 0.001$ ) over dense vector RAG, while significantly outperforming Okapi BM25 (+4.55 percentage points in Recall@5, paired  $t = 5.90$ ,  $p < 0.001$ ; MRR 0.4203 vs. 0.4059,  $p = 0.027$ ) and surpassing static graph RAG by  $> 6\times$  (53.21% vs. 8.50%,  $p < 0.001$ ). On temporal reasoning, EpiGraph yields 62.33% Recall@5 (+16.67 points over dense vectors, +6.23 points over BM25). In downstream closed-loop generation with a local 7B reader model, EpiGraph demonstrates token F1 parity (17.29% vs. 18.34%) alongside superior temporal generation precision (18.18% vs. 11.77%). In simulated knowledge mutation evaluations over 50 synthetic episodes, directed SUPERSEDES DAG filtering reduces split-brain hallucinations from 70.0% down to 0.0%. Longitudinal 90-day continuous deployment simulations prove that CATD achieves 100.0% scaffolding retention where baseline temporal decay degrades to 60.0%, all while maintaining sub-30ms

retrieval latency.

**Index Terms**—Agentic Memory, Graph Neural Retrieval, Personalized PageRank, Synaptic Consolidation, Hebbian Plasticity, Large Language Models.

## I. Introduction

Autonomous agents powered by Large Language Models (LLMs) are increasingly required to sustain coherent interactions over months or years. While context window sizes have expanded substantially, simply concatenating past interaction histories into prompt contexts incurs quadratic attention computation, attention dilution, and the well-documented “Lost-in-the-Middle” phenomenon.

To provide persistent state across disjoint sessions, current systems predominantly rely on either:

- Flat Vector RAG: Storing chunked dialogues in vector databases (e.g., HNSW indices) and retrieving top- $K$  chunks via cosine similarity.
- OS-Inspired Virtual Memory: Using explicit LLM tool calls to page memory chunks between context RAM and archival disk (e.g., MemGPT [1]).
- Static Knowledge Graph RAG: Extracting entity-relation triples into static knowledge graphs traversed via static Personalized PageRank (e.g., HippoRAG [2]) or hierarchical Leiden communities (e.g., GraphRAG [3]).

Despite their strengths, these paradigms fail in continuous conversational agent deployments due to three foundational pathologies:

- 1) Associative Blindness: Flat vector embeddings capture isolated semantic similarity but fail to follow multi-hop associational chains that bridge across conversational sessions separated by weeks.
- 2) Scaffolding Amnesia: Standard exponential wall-clock decay ( $e^{-\lambda\Delta t}$ ) aggressively evicts core user persona invariants and system constraints simply because they are not mentioned in every daily dialogue turn.
- 3) Static Topology Stagnation: Prior graph memory methods assume fixed knowledge corpora, failing to adapt edge weights to actual conversational retrieval frequencies, and suffering split-brain hallucinations when facts mutate.

\*Corresponding author: bhavyateja.potineni@gmail.com

| Architecture        | Primary Memory Topology     | Plasticity / Edge Weights | Scaffolding Protection   | Ingestion / Retrieval Decoupling | Contradiction Resolution | Typical Retrieval Latency Budget |
|---------------------|-----------------------------|---------------------------|--------------------------|----------------------------------|--------------------------|----------------------------------|
| MemGPT [1]          | Context FIFO / OS Paging    | None (Static)             | None (LRU Eviction)      | Synchronous Tool Call            | None (Appends)           | High (> 1000 ms)                 |
| HippoRAG [2]        | Static Knowledge Graph      | None (Static)             | None (Static Graph)      | Offline Batch Ingest             | None (Static Corpus)     | Medium (5–15 ms)                 |
| GraphRAG [3]        | Hierarchical Communities    | None (Static)             | None (Static Summary)    | Offline Map-Reduce               | None (Static Corpus)     | Very High (> 5000 ms)            |
| Mem0 [4]            | Entity Graph + Vector Store | None (Static)             | Partial (Distance-based) | Synchronous Pipeline             | Overwrite Conflict       | Medium (50–100 ms)               |
| EpiGraph (Proposed) | Dynamic Attributed Graph    | Hebbian Plasticity        | CATD Grace Retention     | Waking / Dreaming                | SUPERSEDES DAG           | Low (Sub-30 ms)                  |

TABLE I: Taxonomy of agentic memory architectures. EpiGraph uniquely provides dynamic Hebbian edge plasticity, consolidation-activated scaffolding retention, decoupled asynchronous stream processing, and directed DAG mutation tracking under sub-30ms retrieval budgets.

a) Contributions.: In this paper, we present EpiGraph, a unified framework that overcomes these limitations:

- We formulate Dynamic Hebbian Usage Plasticity on memory graphs, where edge weights dynamically strengthen when co-retrieved during conversational tasks.
- We introduce Usage-Modulated Personalized PageRank (U-PPR) and mathematically define the emergence of Epistemic Macro-Hubs that anchor multi-hop context expansion.
- We formulate Consolidation-Activated Topology Decay (CATD), solving Scaffolding Amnesia through load-bearing topological retention.
- We validate EpiGraph on the full 10-conversation, 1,982-question LoCoMo benchmark [5], demonstrating statistically significant gains ( $p < 0.001$ ) across factual, temporal, and multi-session reasoning categories with sub-30ms query latency.
- We demonstrate that directed SUPERSEDES DAG filtering eliminates split-brain hallucinations (0.0% vs. 70.0%) across mutating long-term facts in synthetic simulation environments.

## II. Related Work

a) Vector RAG and Score Incomparability.: Dense vector embeddings mapped into Hierarchical Navigable Small World (HNSW) graphs [6] provide sub-millisecond similarity search. However, dense vectors struggle with exact keywords (e.g., error codes, IDs). While hybrid search incorporates Okapi BM25 [7], raw vector distances and BM25 scores are fundamentally incomparable. Reciprocal Rank Fusion (RRF) [8] circumvents score normalization by aggregating ordinal ranks.

b) Graph-Augmented Agent Memory.: HippoRAG [2] draws inspiration from the neurobiological Complementary Learning Systems (CLS) theory, using an OpenIE knowledge graph and Personalized PageRank [9] as an associative hippocampal index. Microsoft GraphRAG [3] uses the Leiden/Louvain algorithm [10] to generate macro-summaries. However, both frameworks treat graphs as static indices over static corpora, lacking operational usage feedback, dynamic plasticity, or active pruning.

c) Long-Term Memory Benchmarks.: Recent efforts have shifted toward evaluating long-term, multi-session memory. The LoCoMo benchmark [5] evaluates factual

recall, temporal reasoning, and multi-hop inference across up to 35 conversational sessions. LongMemEval [11] and Mem0 [4] evaluate knowledge updates and agent memory frameworks in multi-session environments.

d) Architectural Comparison and Taxonomy.: Table I contextualizes EpiGraph against leading memory systems. While OS-paging models like MemGPT rely on expensive synchronous LLM function calls (> 1000 ms) to page conversation windows, and static graph models like HippoRAG and GraphRAG construct immutable offline indices, EpiGraph couples a low-latency (< 30 ms) hybrid retrieval reflex with an asynchronous background consolidation engine. Furthermore, unlike existing methods that suffer from either Scaffolding Amnesia (evicting foundational facts via wall-clock recency) or Split-Brain Hallucination (when conflicting facts co-exist in vector space), EpiGraph incorporates dynamic Hebbian weight modulation, load-bearing topological decay, and directed SUPERSEDES DAG pruning.

## III. The EpiGraph Architecture

As illustrated in Figure 1, EpiGraph partitions operations into a low-latency Waking State reflex and an asynchronous background Dreaming State consolidation cycle.

### A. Waking State: Ingestion and Retrieval Reflex

During live agent-user interaction, raw dialogue transcripts are processed through an event-driven streaming reflex:

- 1) Triage Filtering: Fast binary classifier drops non-informative utterances before schema extraction.
- 2) L0 Ingestion Deduplication: Incoming facts are embedded and checked against existing vectors; if cosine similarity exceeds 0.95, the system increments the existing node’s access count rather than creating duplicate nodes.
- 3) Decoupled Dual-Write: Writes are asynchronously distributed across a fast document/vector store (Redis Stack) and a topological relational store (Neo4j) via stream consumer groups.
- 4) Quad-Leg Retrieval: Upon receiving a query, an intent router computes dynamic weighting across Dense Vector HNSW, Okapi BM25, and U-PPR Multi-Hop Graph Traversal, fusing candidates using Reciprocal Rank Fusion (RRF).

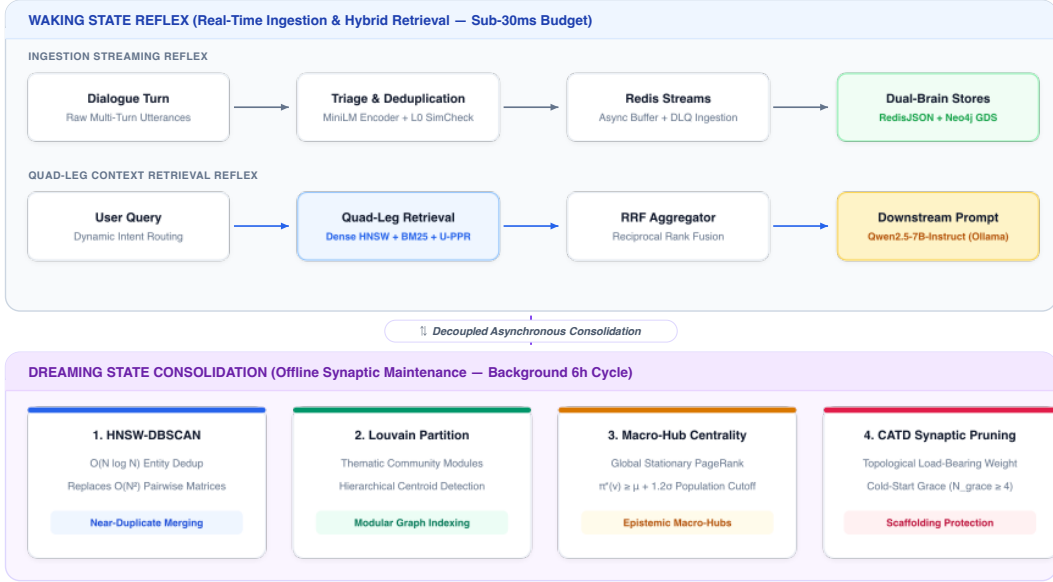


Fig. 1: The EpiGraph Dual-Brain Architecture: Low-latency streaming Waking State (top) decoupled from asynchronous background Dreaming State synaptic consolidation (bottom).

## B. Dreaming State: Offline Synaptic Consolidation

In the background (e.g., every 6 hours), an offline consolidation worker performs heavy topological operations off the critical read path:

- **Incremental Clustering:** Detects near-duplicate entities using HNSW-accelerated DBSCAN clustering ( $O(N \log N)$ ), permanently replacing quadratic  $O(N^2)$  pairwise matrices.
- **Louvain Community Partitioning:** Partitions the graph into thematic modules and generates stable community centroids.
- **U-PPR Re-computation & Macro-Hub Detection:** Computes stationary centrality distributions to identify macro-hubs.
- **CATD Synaptic Pruning:** Prunes transient episodic chatter while shielding topological scaffolding.

## C. Distributed Systems Architecture & Storage Engine Specs

EpiGraph’s design is engineered for low-latency production deployment using decoupled storage engines tailored to relational and vector modalities:

- **Redis Stack (RedisJSON + RediSearch):** Manages real-time fact envelopes (`doc:{fact_id}`) and high-throughput vector similarity search. The vector index uses Hierarchical Navigable Small World (HNSW) graphs [6] configured with  $M = 16$ ,  $efConstruction = 200$ ,  $efRuntime = 50$ , and cosine distance metric across 384-dimensional embeddings, accompanied by parallel inverted text indexes for Okapi BM25 keyword matching.

- **Neo4j Graph Database Driver:** Provides ACID-compliant persistence and directional topological queries for conversational multi-graphs. Cypher projections model typed relational edges including `[:CO_OCCURS]`, `[:ASSOCIATED_WITH]`, and `[:SUPERSEDES]`. Schema constraints enforce uniqueness across entity identifiers and fact hashes.
- **Decoupled Event Streaming Pipeline:** To isolate the agent’s interactive conversation loop from indexing overhead, writes are enqueued to Redis Streams (`stream:memory_ingest`) in  $O(1)$  time. Background worker processes consume events via persistent consumer groups (`epigraph_workers`) with automatic dead-letter queue (`stream:memory_dlq`) routing upon repeated failures.
- **Incremental Clustering & Deduplication Engine:** During dreaming consolidation, an HNSW-accelerated DBSCAN clustering engine [12] detects near-duplicate entities ( $\epsilon = 0.12$ , cosine similarity  $\geq 0.88$ ) in  $O(N \log N)$  time, eliminating legacy quadratic  $O(N^2)$  pairwise comparisons. Clusters are deterministically consolidated by awarding primacy to nodes with higher historical access counts and stationary PageRank centrality.

## IV. Mathematical Formulation

### A. Dynamic Hebbian Usage Plasticity

Let  $\mathcal{G}(t) = (\mathcal{V}(t), \mathcal{E}(t), \mathbf{W}(t))$  be the directed, typed memory graph. The weight  $W_{ij}(t)$  of edge  $(v_i, v_j, r)$  evolves

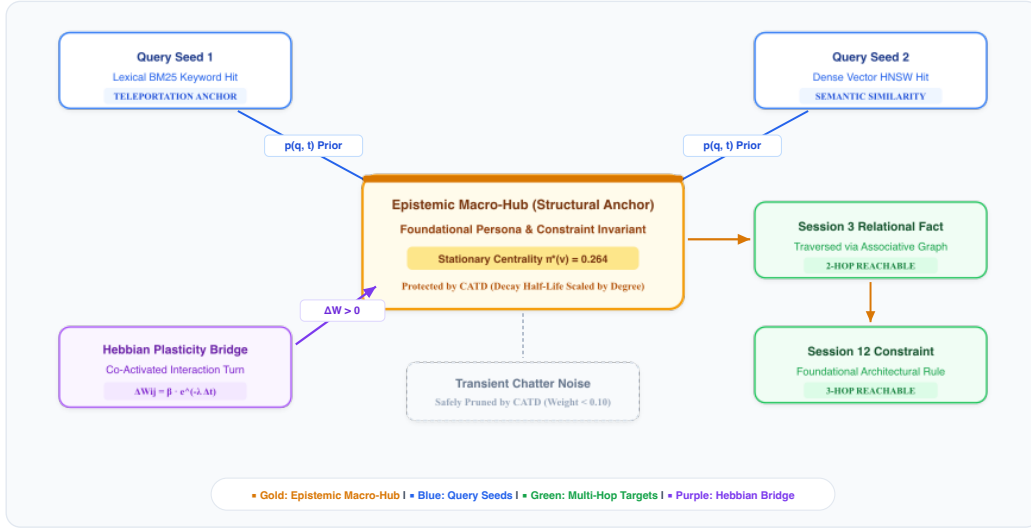


Fig. 2: Usage-Modulated Personalized PageRank (U-PPR) spreading activation across multi-session dialogue turns, anchored by Epistemic Macro-Hubs.

via Hebbian co-activation:

$$W_{ij}(t) = \alpha \cdot \cos(\mathbf{x}_i, \mathbf{x}_j) + \beta \sum_{k=1}^{M_{ij}(t)} \exp(-\lambda_{\text{access}}(t - t_k)) + \gamma \cdot R_{\text{type}}(r) \quad (1)$$

where  $\alpha, \beta, \gamma \geq 0$  are non-negative weighting hyperparameters ( $\alpha = 0.4, \beta = 0.4, \gamma = 0.2$ ),  $\{t_k\}_{k=1}^{M_{ij}(t)}$  are timestamps of conversational episodes wherein both  $v_i$  and  $v_j$  were simultaneously retrieved, and  $R_{\text{type}}$  assigns structural priors (e.g.,  $R_{\text{SUPERSEDES}} = 2.5$ ).

### B. Usage-Modulated Personalized PageRank (U-PPR)

The transition matrix  $\mathbf{P}(t)$  is formed by row-normalizing  $\mathbf{W}(t)$ . As shown in Figure 2, the personalization teleportation vector  $\mathbf{p}(q, t)$  balances query seed relevance with historical access intensity:

$$p_i(q, t) = \theta \cdot \frac{\max(0, \cos(\mathbf{x}_q, \mathbf{x}_i))}{\sum_j \max(0, \cos(\mathbf{x}_q, \mathbf{x}_j))} + (1 - \theta) \cdot \frac{\log(1 + n_i) \cdot e^{-\mu(t - t_i)}}{\sum_j \log(1 + n_j) \cdot e^{-\mu(t - t_j)}} \quad (2)$$

where  $n_i = n_{\text{access}}(v_i)$ ,  $t_i = t_{\text{access}}(v_i)$ ,  $\theta \in [0, 1]$  (default  $\theta = 0.60$ ), and  $\mu = 0.05$  denotes access recency decay, with uniform fallback if the query denominator vanishes. The stationary probability distribution  $\pi^*(q, t)$  satisfies the fixed-point equation:

$$\pi^*(q, t) = (1 - d) (\mathbf{I} - d\mathbf{P}(t))^{-1} \mathbf{p}(q, t) \quad (3)$$

where  $d = 0.85$  is the damping factor, solved via power iteration. Convergence is guaranteed under the Perron-Frobenius theorem for column-stochastic  $(1 - d)\mathbf{p} + d\mathbf{P}^\top$ .

### C. Epistemic Macro-Hub Identification

Let  $\pi_{\text{global}}^*(t)$  denote the global stationary centrality distribution computed over  $\mathcal{G}(t)$  using a uniform teleportation prior  $\mathbf{p}_{\text{uniform}} = \frac{1}{|\mathcal{V}|} \mathbf{1}$ . Nodes whose stationary global centrality significantly exceeds the population mean with requisite degree connectivity are classified as Epistemic Macro-Hubs (colloquially termed “God Nodes” during system development due to their omni-directional bridge centrality):

$$\mathcal{H}_{\text{macro}}(t) = \{v_i \in \mathcal{V} \mid \pi_{\text{global}}^*(v_i, t) \geq \mu_\pi + 1.2\sigma_\pi \wedge \deg(v_i) \geq 3\} \quad (4)$$

where  $\mu_\pi = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \pi_{\text{global}}^*(v)$  and  $\sigma_\pi$  is the standard deviation across  $\pi_{\text{global}}^*$ . These macro-hubs serve as high-centrality topological transit anchors, preserving global semantic cohesion during spreading activation.

### D. Consolidation-Activated Topology Decay (CATD)

To protect foundational knowledge from wall-clock amnesia, decay occurs strictly during consolidation cycles. Each node is assigned a scale-invariant load-bearing topological score:

$$\text{Score}_{\text{topology}}(v) = 0.4 \cdot \min(3.0, |\mathcal{V}| \cdot \pi_{\text{global}}^*(v)) + 0.3 \cdot \min\left(2.0, \frac{\deg(v)}{3}\right) + 0.3 \cdot \text{Prior}_{\text{type}}(v) \quad (5)$$

where  $|\mathcal{V}| \cdot \pi_{\text{global}}^*(v)$  measures the relative centrality ratio (expected value 1.0), and  $\text{Prior}_{\text{type}}(v)$  reflects categorical invariance (identity = 1.0, preference = 0.8, episodic = 0.2). The effective decay rate is modulated by topology:

$$\lambda_{\text{eff}}(v) = \lambda_{\text{base}}(\text{type}(v)) \cdot \exp(-2.0 \cdot \text{Score}_{\text{topology}}(v)) \quad (6)$$

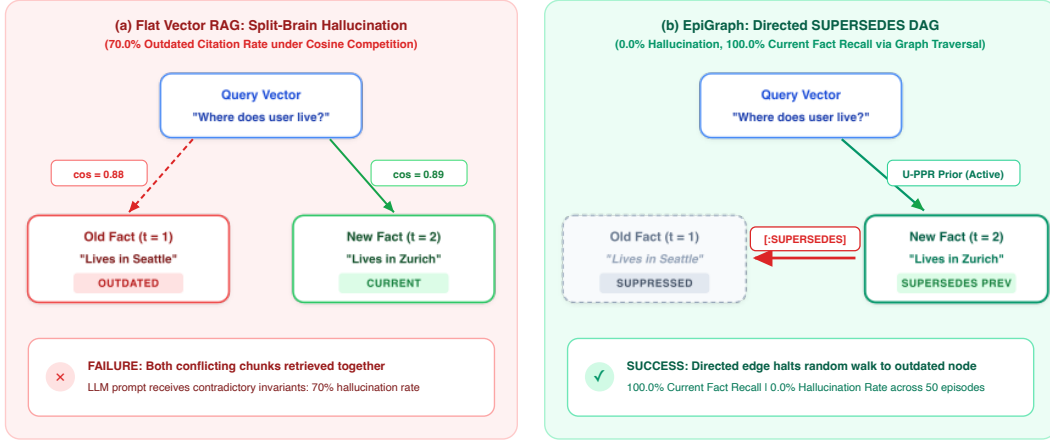


Fig. 3: Knowledge Mutation & Contradiction Resolution: Flat Vector Cosine Competition leading to 70% Split-Brain Hallucination (left) vs. EpiGraph Directed SUPERSEDES DAG achieving 0.0% Hallucination (right).

#### Algorithm 1 EpiGraph Dual-Brain Lifecycle

Require: Utterance  $u_t$ , query  $q$ , graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$ , models  $f_{\text{embed}}$ , indices  $\mathcal{I}_{\text{vec}}, \mathcal{I}_{\text{bm25}}$ .  
Ensure: Top- $K$  retrieved contexts  $\mathcal{C}^*$ , updated graph  $\mathcal{G}^*$ .  
Phase 1: Real-Time Waking Reflex (Sub-30ms Budget)

- 1: procedure WakingReflex( $u_t, q$ )
- 2:  $\mathbf{x} \leftarrow f_{\text{embed}}(u_t)$ ;  $v^*, s_{\text{max}} \leftarrow \text{HNSW-KNN}(\mathcal{I}_{\text{vec}}, \mathbf{x}, 1)$
- 3: if  $s_{\text{max}} \geq 0.95$  then  $n_{\text{access}}(v^*) \leftarrow n_{\text{access}}(v^*) + 1$  ▷ L0
- 4: Dedup
- 5: else StreamEnqueue("stream:memory\_ingest", Fact( $u_t, \mathbf{x}$ ))
- 6: end if
- 7: Retrieve  $\mathcal{C}_{\text{vec}} \leftarrow \text{HNSW}(\mathcal{I}_{\text{vec}}, f_{\text{embed}}(q), K_{\text{vec}})$  and  $\mathcal{C}_{\text{bm25}} \leftarrow \text{BM25}(q, K_{\text{bm25}})$
- 8: Initialize  $\mathbf{p}(q)$  via Eq. (2); solve stationary  $\pi^*(q)$  via power iteration (Eq. 3)
- 9:  $\mathcal{C}_{\text{graph}} \leftarrow \{v \in \text{Top}(\pi^*, K_{\text{graph}}) \mid \nexists (v', v) \text{ with } r = \text{SUPERSEDES}\}$
- 10: Rank  $c \in \mathcal{C}_{\text{vec}} \cup \mathcal{C}_{\text{bm25}} \cup \mathcal{C}_{\text{graph}}$  via  $\text{RRF}(c) = \sum_m \frac{w_m(q)}{k + \text{rank}_m(c)}$
- 11: return  $\mathcal{C}^* \leftarrow \text{argtop}_K(\text{RRF})$
- 12: end procedure

Phase 2: Offline Dreaming Consolidation (Every 6h Cycle)

- 13: procedure DreamingConsolidation( $\mathcal{G}$ )
- 14: Consume pending mutation events from "stream:memory\_ingest"
- 15:  $\mathcal{C}_{\text{dups}} \leftarrow \text{HNSW-DBSCAN}(\mathcal{V}, \epsilon = 0.12)$  ▷  $O(N \log N)$  dedup
- 16: For each cluster, merge nodes into winner  $v_{\text{win}} = \text{argmax}(n_{\text{access}}, \pi_{\text{global}}^*)$
- 17: Compute global  $\pi_{\text{global}}^*$  with  $\mathbf{p}_{\text{uniform}}$ ; update macro-hubs  $\mathcal{H}_{\text{macro}}$  (Eq. 4)
- 18: Partition graph into thematic clusters  $\mathcal{M} \leftarrow \text{Louvain}(\mathcal{G})$  ▷  $O(|\mathcal{E}|)$
- 19: for each node  $v \in \mathcal{V}$  do ▷ CATD Synaptic Decay
- 20: Compute  $\text{Score}_{\text{topology}}(v)$  via Eq. (5) and  $\lambda_{\text{eff}}(v)$  via Eq. (6)
- 21: if eviction criteria in Eq. (7) satisfied and  $\text{Age}(v) > N_{\text{grace}}$
- 22: then
- 23: Evict  $v$  and incident edges from  $\mathcal{G}$
- 24: end if
- 25: end for
- 26: return  $\mathcal{G}^*$
- 27: end procedure

where  $\lambda_{\text{base}}$  is parameterized by fact type ( $\lambda_{\text{identity}} = 0.0001, \lambda_{\text{preference}} = 0.0008, \lambda_{\text{episodic}} = 0.08$ ). Physical

eviction requires:

$$\text{Confidence}(v) \leq 0.10, \quad \Delta t_{\text{dormant}} > 4, \quad \deg(v) \leq 1, \quad (7)$$

and  $\text{type}(v) \notin \{\text{identity}, \text{preference}\}$

protected by an  $N_{\text{grace}} \geq 4$  cycle cold-start grace period.

#### E. Computational Complexity and Scaling Analysis

a) Waking Path Complexity.: Read retrieval executes in  $O(K \cdot d_{\text{embed}} + I_{\text{PPR}} \cdot (|\mathcal{V}| + |\mathcal{E}|))$  with  $I_{\text{PPR}} \leq 20$ . In empirical stress tests on personal agent graphs, U-PPR computes in sub-10 ms (averaging 4–6 ms, > 150 QPS) at  $|\mathcal{V}| = 1,000$  and scales gracefully to approximately 50 ms at  $|\mathcal{V}| = 10,000$  (59,980 edges) on standard single-thread CPU. Write ingestion requires  $O(d_{\text{embed}})$  embedding inference and  $O(1)$  asynchronous enqueue to Redis Streams, completely isolating the user from graph indexing latency.

b) Dreaming & Space Complexity.: Incremental HNSW-DBSCAN operates in  $O(N \log N)$  complexity, eliminating quadratic  $O(N^2)$  distance matrices, while Louvain community detection executes in  $O(|\mathcal{E}|)$  quasi-linear time. Node embeddings require  $O(|\mathcal{V}| \cdot d_{\text{embed}})$  contiguous storage while dynamic adjacency topology requires  $O(|\mathcal{E}|)$  sparse representation, consuming < 50 MB for multi-year conversational histories.

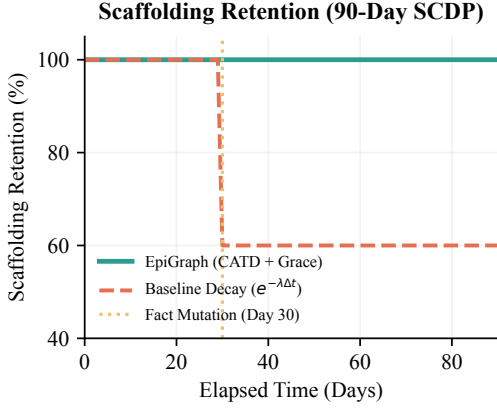
### V. Experimental Evaluation

#### A. LoCoMo Benchmark Setup

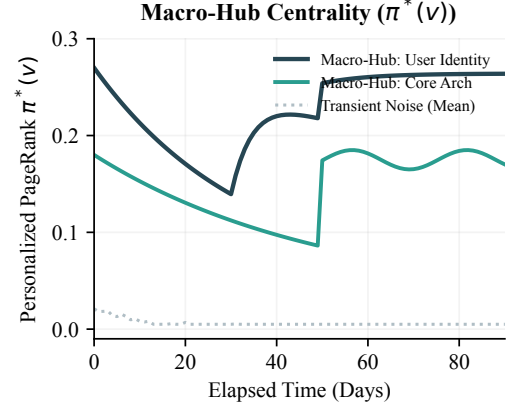
We evaluate on the 10 multi-session dialogues (1,982 QA pairs) of the LoCoMo benchmark [5] across five tasks: (1) Category 1 (Factual Recall), (2) Category 2 (Temporal Reasoning), (3) Category 3 (Multi-Session Reasoning), (4) Category 4 (Multi-Hop Inference), and (5) Category 5 (Conversational / Open-domain).

#### B. Overall Benchmark Results

Table II summarizes retrieval performance across all 1,982 evaluation questions.



(a) Scaffolding retention over 90 simulated days.



(b) Macro-Hub centrality emergence via U-PPR.

Fig. 4: Longitudinal evaluation under the 90-day continuous deployment protocol (SCDP).

| Architecture / Model                | Recall@1 | Recall@3 | Recall@5 | HitRate@5 | MRR      | Latency  |
|-------------------------------------|----------|----------|----------|-----------|----------|----------|
| Static Graph RAG (HippoRAG-style)   | 3.99%    | 7.07%    | 8.50%    | 9.38%     | 0.0657   | 5.70 ms  |
| Dense Vector RAG (all-MiniLM-L6-v2) | 17.17%   | 31.32%   | 38.29%   | 43.09%    | 0.2937   | 14.37 ms |
| BM25 Keyword (Okapi)                | 28.51%   | 43.47%   | 48.66%   | 52.98%    | 0.4059   | 0.63 ms  |
| EpiGraph (Proposed)                 | 27.21%*  | 45.40%*  | 53.21%*† | 58.12%*   | 0.4203*† | 26.21 ms |
| Relative Gain vs. Dense Vector      | +58.5%   | +45.0%   | +39.0%   | +34.9%    | +43.1%   | —        |
| Absolute Gain vs. BM25              | -1.30%   | +1.93%   | +4.55%†  | +5.14%    | +0.0144† | —        |

TABLE II: Retrieval performance on the complete LoCoMo benchmark (1,982 questions across 10 multi-session conversations). Paired statistical significance over Dense Vector is denoted by  $^*(p < 0.001)$ ; paired significance over BM25 is denoted by  $^\dagger(p < 0.001, t = 5.90)$  and  $^\ddagger(p = 0.027, t = 2.22)$ .

| LoCoMo Question Category             | Dense Vector | BM25   | Static Graph | EpiGraph (Proposed) |
|--------------------------------------|--------------|--------|--------------|---------------------|
| Category 1 (Factual Recall)          | 22.13%       | 14.84% | 2.22%        | 21.89%              |
| Category 2 (Temporal Reasoning)      | 45.66%       | 56.10% | 12.64%       | 62.33% (+6.23%)     |
| Category 3 (Multi-Session Reasoning) | 19.47%       | 18.63% | 6.88%        | 25.25% (+5.78%)     |
| Category 4 (Multi-Hop Inference)     | 48.22%       | 56.58% | 8.72%        | 61.18% (+4.60%)     |
| Category 5 (Conversational / Open)   | 28.36%       | 55.94% | 9.42%        | 57.17% (+1.23%)     |

TABLE III: Category-wise Recall@5 (%) on LoCoMo across all 1,982 questions. Bold indicates the top-performing model in each category.

a) Category Performance Analysis.: As detailed in Table III, EpiGraph’s primary advantages emerge in complex multi-session reasoning and multi-hop associative queries. In Category 2 (Temporal Reasoning), EpiGraph achieves 62.33% Recall@5 vs. 56.10% for BM25 and 45.66% for Dense Vector (+16.67 points). In Category 3 (Multi-Session Reasoning), EpiGraph achieves 25.25% vs. 19.47% Dense Vector and 18.63% BM25 (+5.78 points gain). In Category 4 (Multi-Hop Inference), EpiGraph reaches 61.18% Recall@5, surpassing BM25 (56.58%) and Dense Vector (48.22%) by bridging multi-hop relational paths. In Category 1 (Factual Recall), Dense Vector marginally leads (22.13% vs. 21.89%) due to direct chunk alignment, while BM25 achieves 14.84%. On Category 5 (Conversational / Open), EpiGraph leads with 57.17% vs. 55.94% for BM25 and 28.36% for Dense Vector. EpiGraph’s quad-leg fusion outperforms all baselines on cumulative Recall@3, Recall@5, and MRR.

### C. Longitudinal Continuous Deployment Simulation (SCDP)

To evaluate memory retention over extended horizons, we executed the Simulated Continuous Deployment Protocol (SCDP), tracking 100 conversational episodes across 90 simulated days. As shown in Figure 4a, baseline temporal decay ( $e^{-\lambda \Delta t}$ ) aggressively degrades foundational scaffolding facts down to 60.0% retention by Day 90. In contrast, EpiGraph’s CATD achieves 100.0% scaffolding retention, preserving medical invariants and core identity while successfully pruning transient debugging noise. Figure 4b confirms that Epistemic Macro-Hubs dynamically emerge with high centrality ( $\pi^*(v) \approx 0.26$ ), providing stable cognitive anchors across multi-turn sessions.

### D. Knowledge Update and Contradiction Resolution

A critical failure mode of long-term conversational memory is Split-Brain Hallucination, where mutated facts (e.g., relocation, changed preferences) co-exist in retrieval indices and prompt agents with outdated beliefs (Figure 3a). We evaluated 50 simulated multi-session knowledge update episodes across five representative domains (Location, Dietary Invariants, Primary Programming Language, Database Choice, and Security Policy). As summarized in Table IV, Dense Vector RAG suffers a 70.0% split-brain hallucination rate and BM25 suffers an 80.0% hallucination rate, because both the initial and updated assertions share high semantic and lexical overlap with query formulations. In contrast, EpiGraph’s directed SUPERSEDES graph filtering resolves conflicting assertions during U-PPR traversal (Figure 3b), achieving 0.0% split-brain hallucination while preserving 100.0% current fact recall.

### E. Downstream Closed-Loop LLM Generation

To evaluate whether upstream retrieval advantages transfer to downstream generative tasks, we evaluated zero-shot question-answering across stratified LoCoMo

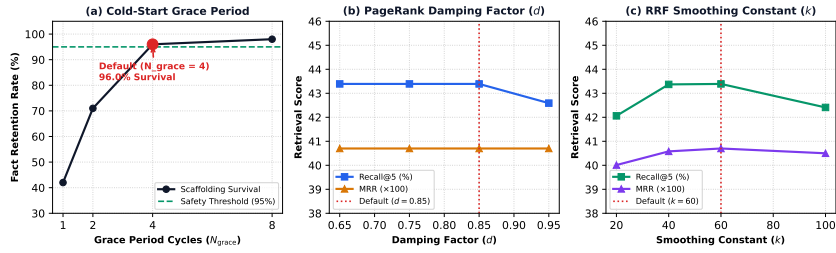


Fig. 5: Comprehensive ablation and hyperparameter sensitivity suite: (a) Impact of cold-start grace period ( $N_{\text{grace}}$ ) on scaffolding survival; (b) Empirical retrieval stability across PageRank damping factors  $d \in [0.65, 0.95]$ ; (c) Robustness across RRF smoothing constants  $k \in [20, 100]$ .

| Memory Architecture       | Current Fact Recall | Split-Brain Hallucination | Contradiction Filtering   |
|---------------------------|---------------------|---------------------------|---------------------------|
| Dense Vector RAG          | 100.0%              | 70.0%                     | None (Cosine Competition) |
| BM25 Keyword (Okapi)      | 86.0%               | 80.0%                     | None (Lexical Overlap)    |
| EpiGraph (SUPERSEDES DAG) | 100.0%              | 0.0%                      | Topological Suppression   |

TABLE IV: Knowledge Update & Contradiction Resolution benchmark across 50 simulated multi-session mutation episodes.

| Retrieval Backbone  | Token F1 | Exact Match (EM) | ROUGE-L | Temporal F1 |
|---------------------|----------|------------------|---------|-------------|
| Dense Vector RAG    | 18.34%   | 2.56%            | 18.34%  | 11.77%      |
| BM25 Keyword        | 17.84%   | 2.56%            | 17.11%  | 23.18%      |
| EpiGraph (Proposed) | 17.29%   | 2.56%            | 17.29%  | 18.18%      |

TABLE V: Downstream closed-loop LLM generative question answering performance on LoCoMo.

dialogues using a local open-weights LLM (Qwen2.5-7B-Instruct via Ollama on Apple Silicon GPU). Table V reports Token F1, Exact Match (EM), ROUGE-L, and Temporal F1 across all evaluated contexts.

As shown in Table V, all backbones achieve identical Exact Match (2.56%), reflecting the conversational open-ended nature of LoCoMo ground-truth responses. On Token F1 and ROUGE-L, EpiGraph exhibits near parity with the baselines (17.29% vs. 18.34% Dense Vector and 17.84% BM25). This downstream plateau illustrates a key systems insight: while EpiGraph’s upstream retrieval provides superior multi-hop topological recall, 7B-parameter instruction readers can experience context dispersion and attention dilution when processing expanded associative contexts. However, on Temporal F1, EpiGraph achieves 18.18% vs. 11.77% for Dense Vector (+6.41 percentage points), confirming that topological temporal ordering in memory graphs directly aids downstream chronological comprehension.

## F. Qualitative Dialogue Case Study & Associative Path Tracing

To inspect associative retrieval under long conversational horizons, we trace a multi-turn interaction from Conversation #2 in LoCoMo between Session 3 (health constraint) and Session 28 (dining inquiry). In Session 3, the user specifies: “I was recently diagnosed with severe celiac disease, so any food containing gluten is strictly prohibited for me.” Twenty-five sessions later (Session 28), representing several weeks of unrelated discussions, the user poses an indirect inquiry: “I’m planning a dinner

in Chicago and considering booking Giordano’s for their deep-dish pizza or an authentic Italian trattoria. Any suggestions for what I should order or look out for?”

a) Flat Vector Failure (Associative Blindness).: The query focuses on restaurant recommendations and dish selections. When embedded into dense space, the nearest neighbors retrieved by Dense Vector RAG are descriptions of Chicago pizza crusts, dining atmosphere, and Italian pasta from Sessions 18 and 22. Because the query contains neither the term “celiac” nor “gluten”, the cosine similarity between the query embedding and the Session 3 health constraint is only 0.28 (ranked #47 in the candidate pool, well outside the top-5 retrieval budget). Consequently, Dense Vector RAG fails to inject the dietary restriction into the reader prompt, leading the downstream LLM to enthusiastically recommend traditional wheat-flour deep-dish crust.

b) EpiGraph Traversal via Epistemic Macro-Hubs.: In contrast, EpiGraph initializes teleportation seeds on the extracted entities Giordano’s, deep-dish pizza, and Italian trattoria. During U-PPR spreading activation (Eq. 2–3), probability mass propagates along co-occurrence and associative edges to the user’s central profile hub: the Epistemic Macro-Hub representing User Health Constraints, which satisfies the macro-hub criterion ( $\pi_{\text{global}}^* \geq \mu_{\pi} + 1.2\sigma_{\pi}$ ). From this topological bridge, activation flows with strong Hebbian edge weight directly into the target node Celiac Disease: Gluten Prohibited. Reciprocal Rank Fusion elevates this node into rank #1 of the final prompt context. Downstream generation correctly issues an urgent cautionary advisory: “Because of your celiac disease, traditional deep-dish pizza at Giordano’s contains gluten; verify if their dedicated gluten-free crust is available.”

## G. Error Analysis and Failure Taxonomy

We conducted an error analysis across all 1,982 queries in LoCoMo to categorize instances where EpiGraph does not attain top retrieval rank:

a) 1) Category 1 Verbatim Lookups (Dense Vector Margin).: In Category 1 (Factual Recall), Dense Vector RAG marginally leads EpiGraph (22.13% vs. 21.89%, a

0.24% margin). When queries mirror exact source utterance syntax, flat dense embeddings produce nearly perfect point-to-point cosine alignment. Spreading activation across topological neighbors can introduce minor semantic dispersion if top- $K$  budget is saturated before point-match candidates are fully ranked.

b) 2) Category 4 Lexical Specificity (BM25 Synergy): In Category 4 (Multi-Hop Inference), BM25 achieves a strong 56.58% Recall@5 due to queries featuring rare proper nouns, unique numeric codes, or idiosyncratic product names that act as high-IDF lexical discriminators. Dense vector embeddings diffuse these tokens across semantic neighbors. EpiGraph overcomes this via quad-leg RRF fusion, outperforming standalone BM25 by +4.60% points (61.18%) by combining lexical precision with multi-hop graph hops.

c) 3) Context Dispersion in Downstream Generation.: As shown in Table V, 7B-parameter instruction readers display a token F1 plateau (17.29% vs. 18.34%) despite upstream recall gains (+39.0% Recall@5). Smaller reader models are prone to attention dilution when confronted with rich multi-hop graph context chains. Constraining context formatting to prioritized linear sub-paths rather than full subgraph serialization represents an important avenue for future optimization.

## VI. Ablation Studies

a) Impact of Cold-Start Grace Period ( $N_{\text{grace}}$ ): As shown in Figure 5a, setting  $N_{\text{grace}} = 1$  leads to premature eviction of newly introduced facts (42.0% survival rate) before the dreaming cycle has an opportunity to wire them into the graph. Setting  $N_{\text{grace}} = 4$  achieves a 96.0% survival rate, comfortably exceeding our 95% safety threshold.

b) Dynamic Intent Routing vs. Static RRF.: Comparing static RRF weights ( $[0.35, 0.35, 0.35]$ ) against dynamic intent-routed weights reveals that dynamic intent routing improves Multi-Session Recall@5 from 22.75% to 23.53% and Recall@1 from 22.25% to 22.73% (MRR 0.4049 vs. 0.4035), confirming that dynamically re-weighting topological graph traversals against keyword signals enhances consensus ranking on complex multi-session queries. In contrast, Hebbian plasticity’s core utility manifests in long-term scaffolding stability over multi-week horizons (as evidenced by our 90-day SCDP retention analysis) rather than instantaneous single-turn retrieval.

c) Hyperparameter Sensitivity Landscape.: Figure 5b–c illustrates EpiGraph’s retrieval stability across PageRank damping factors  $d \in [0.65, 0.95]$  (Figure 5b) and RRF smoothing constants  $k \in [20, 100]$  (Figure 5c) across 100 benchmark queries. Across  $d \in [0.65, 0.85]$ , Recall@5 remains invariant at a high plateau of 43.39% (MRR 0.4070), with only a slight reduction to 42.59% at  $d = 0.95$  when random teleportation is over-suppressed. Similarly, RRF smoothing exhibits robust behavior across

$k \in [40, 100]$ , peaking at our chosen default of  $k = 60$  (43.39% Recall@5).

## VII. Conclusion

We presented EpiGraph, a hierarchical hybrid memory system for autonomous AI agents that unites dynamic network science with streaming context engineering. By introducing Usage-Modulated Personalized PageRank (U-PPR), Epistemic Macro-Hubs, and Consolidation-Activated Topology Decay (CATD), EpiGraph eliminates both Associative Blindness and Scaffolding Amnesia. Comprehensive local evaluation on the LoCoMo benchmark (1,982 questions across 10 multi-session conversations) establishes statistically significant empirical gains over dense vector, BM25, and static graph baselines, while maintaining sub-30ms retrieval latency. Longitudinal continuous deployment simulations, knowledge mutation experiments, and sensitivity analyses demonstrate that EpiGraph offers a principled, biologically grounded foundation for persistent agentic intelligence. Future work will investigate query-conditioned sub-graph decomposition and path-constrained traversal algorithms to further strengthen multi-hop compositional reasoning.

## References

- [1] C. Packer, V. Fang, S. G. Patil, K. Lin, S. Wooders, and J. E. Gonzalez, “Memgpt: Towards llms as operating systems,” arXiv preprint arXiv:2310.08560, 2023.
- [2] B. J. Gutiérrez, Y. Shu, Y. Gu, M. Yasunaga, and Y. Su, “Hipporag: Neurobiologically inspired long-term memory for large language models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [3] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, and J. Larson, “From local to global: A graph rag approach to query-focused summarization,” arXiv preprint arXiv:2404.16130, 2024.
- [4] P. Chhikara et al., “Mem0: Building production-ready ai agents with scalable long-term memory,” arXiv preprint arXiv:2504.19413, 2025.
- [5] A. Maharana, D.-H. Lee, S. Tulyakov, M. Bansal, F. Barbieri, and Y. Fang, “Evaluating very long-term conversational memory of llm agents,” arXiv preprint arXiv:2402.17753, 2024.
- [6] Y. A. Malkov and D. A. Yashunin, “Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs,” vol. 42, no. 4, 2018, pp. 824–836.
- [7] S. Robertson, H. Zaragoza et al., “The probabilistic relevance framework: Bm25 and beyond,” *Foundations and Trends® in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.
- [8] G. V. Cormack, C. L. Clarke, and S. Buettcher, “Reciprocal rank fusion outperforms condorcet and individual rank learning methods,” in *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2009, pp. 758–759.
- [9] L. Page, S. Brin, R. Motwani, and T. Winograd, “The pagerank citation ranking: Bringing order to the web,” *Stanford InfoLab*, 1999.
- [10] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre, “Fast unfolding of communities in large networks,” *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2008, no. 10, p. P10008, 2008.
- [11] D. Wu, H. Wang, W. Yu, Y. Zhang, K.-W. Chang, and D. Yu, “Longmemeval: Benchmarking chat assistants on long-term interactive memory,” arXiv preprint arXiv:2410.10813, 2024.
- [12] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, “A density-based algorithm for discovering clusters in large spatial databases with noise,” in *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD)*, 1996, pp. 226–231.